Knowledge Check

Table of Contents

1. What is an operating system? Provide with example. 2

2. What are the three main purposes of an operating system? List them and explain briefly. 2

3. Name three types of operating systems and give one example for each. 2

4. What is the kernel?How is it different from the full operating system? 2

5. What is a process? How is it different from a program? 2

6. What is virtualization? Name one benefit of using virtual machines. 3

7. Name two computing environments where operating systems are used 3

1. What is an operating system? Provide with example.

Ans: An operating system is software that manages a computer's hardware. It also provides a basis for application programs and acts as an intermediary between the computer user and the computer hardware. Eg: Linux, macOS, Windows etc.

2. What are the three main purposes of an operating system? List them and explain briefly.

Ans:

- Acts as a bridge: It serves as the middleman between the user and the computer hardware.
- Allocates resources: It organizes and distributes hardware resources—like the CPU, memory, and I/O devices—to different programs.
- Provides a base: It creates a stable, secure foundation for application programs to run properly.

3. Name three types of operating systems and give one example for each.

- Ans:
- Desktop OS – Example: Microsoft Windows
- Mobile OS – Example: Android
- Open-Source OS – Example: Linux

4. What is the kernel?How is it different from the full operating system?

Ans:

- The Kernel: The core program of the operating system that runs at all times. It directly controls the hardware and handles essential tasks like memory management.
- The Difference: The kernel is just the core engine. The full operating system includes the kernel plus everything else you interact with, such as system programs (file managers, settings) and user interfaces.

5. What is a process? How is it different from a program?

Ans:

- Program: A passive collection of code stored on your hard drive (like a file or an installer). It is not doing anything until you open it.
- Process: An active program in execution. When you run a program, it loads into memory and becomes a process that actively uses the CPU.

6. What is virtualization? Name one benefit of using virtual machines.

Ans:

- Virtualization: A technology that lets you split a single physical computer's hardware into multiple isolated virtual computers (called Virtual Machines or VMs).
- One Benefit: Resource Efficiency. It allows you to run multiple different operating systems (like Windows and Linux) on the exact same physical machine at the same time, saving hardware costs.

7. Name two computing environments where operating systems are used

Ans:

- Mobile Computing Environment: Used on portable devices like smartphones and tablets.
- Cloud / Distributed Computing Environment: Used on massive networks of servers that host websites and apps over the internet.